

Project Planning

Project: PocketSmart AI – Your Smart Budget & Recommendation Assistant

1. Project Objective

Develop and deploy a GenAI-powered cross-platform recommendation system that provides budget-based suggestions for home decor, party planning, and jewelry.

2. Major Development Activities

- Requirement analysis and identification of target user needs.
- Design of the problem-solution fit, proposed solution, and solution architecture.
- Development of Home, Party, and Jewelry planner workflows.
- Integration of Gemini 1.5 Flash Pro for AI-generated recommendations.
- Implementation of FastAPI backend routes and recommendation history.
- Development of responsive frontend pages and recommendation displays.
- Testing of all three planners, authentication, and recommendation history.
- Deployment and demonstration of the completed application.

3. Planned Modules

Home Planner – budget, room type, and quantity based recommendations.

Party Planner – budget, guest count, event type, venue, and budget allocation for catering, decoration, and entertainment.

Jewelry Planner – budget, occasion, style, and optional outfit image based recommendations.

Recommendation History – access to previously generated recommendation sessions and details.

4. Backend Plan

FastAPI backend with Uvicorn deployment and routes including /generate-home, /generate-party, /generate-jewelry, authentication/session routes, /recommendations-details, /history, and /startup.

5. Frontend Plan

Responsive pages for Home, Testimonials, Register, Login, Dashboard, planner pages, recommendation pages, and recommendation history.

6. Testing and Validation Plan

- Test Home Planner with valid budget and room inputs.
- Test Party Planner with budget, guest count, event type, and venue.
- Test Jewelry Planner with budget, occasion, and style.
- Verify recommendation details and history.
- Verify authentication and session-related flows.

7. Deployment Plan

Deploy the FastAPI application as a web service using Uvicorn and validate the live application through end-to-end planner workflows.

8. Final Deliverables

Working PocketSmart AI web application, tested planner workflows, recommendation history, project documentation, GitHub repository, and project demonstration.